Lambdas and Callables

Function pointers and functors first, then how a lambda works, captures and their lifetime traps, init-capture, generic lambdas, and when to use a function pointer, functor, lambda or `std::function`.

Before lambdas: function pointers and functors

Algorithms such as `std::sort` and `std::count_if` take "something callable". Before C++11 there were two ways to supply it, and a lambda is a shorter way to write the second.

A function pointer

```
bool descending(int a, int b) { return a > b; }
int twice(int x) { return 2 * x; }
void apply(int (*f)(int), int x) { std::cout << f(x) << '\n'; }   // a parameter that points to a function

std::vector<int> v{5, 2, 9, 1};
std::sort(v.begin(), v.end(), descending);          // 9 5 2 1: pass the function by name
apply(twice, 21);                                   // 42
```

- A function pointer is just an address (8 bytes). It can point to any function with the **same signature**, chosen at run time.
- It has **no state**: the function sees only its arguments and globals. To give `descending` a limit you would need a global variable.
- The call goes through the pointer, so the compiler usually cannot inline it.

A functor (function object)

A **functor** is a class with an `operator()`. An object of it can be called like a function, and it can **carry state** in its members.

```
struct GreaterThan {
    int limit;                                          // the state
    bool operator()(int x) const { return x > limit; }
};
GreaterThan gt{4};
gt(5);                                                  // true
gt(3);                                                  // false
std::count_if(v.begin(), v.end(), GreaterThan{1});      // 3: how many elements
are above 1
```

- Each functor class is its own type, so a template like `std::count_if` knows exactly which `operator()` to call and can **inline** it.
- The cost: a whole class, written away from the place where it is used, for one small piece of logic.

SO WHAT IS A LAMBDA? A lambda is a functor the compiler writes for you, in place: `[limit](int x) { return x > limit; }` is the `GreaterThan` above, with `limit` as its member. The next sections show how it works and where it goes wrong.

READ MORE

C++ Primer (Lippman, Lajoie, Moo)

- 6.7 Pointers to functions (PDF pages 276 to 281)

Programming: Principles and Practice (Stroustrup)

- 21.1 to 21.2, 21.5 Standard algorithms, function objects, sorting and searching (PDF pages 617 to 626 and 633 to 635)

Effective Modern C++ (Meyers)

- Item 31, Avoid default capture modes (PDF pages 234 to 242)
- Item 32, Use init capture to move objects into closures (PDF pages 242 to 247)

What a lambda is



Capture by reference stores a reference instead; it dangles if the lambda outlives the variable.

A lambda is an unnamed function object written in place: the compiler generates a small class whose constructor stores the captures and whose `operator()` holds the body.

```
//  [captures] (parameters) -> return_type { body }
auto add = [](int a, int b) { return a + b; };
add(2, 3);                                          // 5

std::vector<int> v{5, 2, 9, 1};
std::sort(v.begin(), v.end(), [](int a, int b) { return a > b; });    //
descending: an inline comparator
```

- Every lambda has its **own unique type**, so store it with `auto` (or a template parameter).
- A lambda with **no captures** converts to a plain function pointer. Compiled: `sizeof` of such a lambda is 1, of one capturing an `int` is 4, of a function pointer 8.
- `operator()` is `const` by default; `mutable` lifts that (next sections).

READ MORE

C++ Primer (Lippman, Lajoie, Moo)

- 10.3 Customizing operations (PDF pages 414 to 429)

Captures: by value and by reference

```
int factor = 3;
auto byValue = [factor](int x) { return x * factor; };    // copies factor
now
auto byRef   = [&factor](int x) { return x * factor; };   // refers to the
live variable
auto all     = [=] (int x) { return x * factor; };        // [=] all by
value, [&] all by reference
```

▼ PREDICT **Capture by value vs by reference**

```
int x = 1;
auto byVal = [x]  { return x; };
auto byRef = [&x] { return x; };
x = 2;
std::cout << byVal() << byRef();
```

> 12

`byVal` took a snapshot of `x` (1) when it was created. `byRef` sees the variable itself, so it reads the new value 2. A by-value capture is *const* inside the lambda unless you add `mutable`.

Step through the two capture modes:

STEP THROUGH **Lambda captures: by value vs by reference**

```
1  int x = 1;
2  auto byVal = [x]  { return x; };
3  auto byRef = [&x] { return x; };
4  x = 2;
5  std::cout << byVal() << byRef();
```

STEP	RUNS	WHAT HAPPENS	MEMORY AFTER (CHANGED IN BOLD)
1	line 1 `int x = 1;`	A local **x** holds 1.	**`x = 1`**
2	line 2 `auto byVal = [x] { return x; };`	`[x]`: the lambda is a small object with its own **member copy** of x, taken *now*.	`x = 1` | **`byVal.x = 1`**
3	line 3 `auto byRef = [&x] { return x; };`	`[&x]`: the lambda stores a **reference** to the live x. Follow the arrow.	`x = 1` | `byVal.x = 1` | **`byRef.x = alias`**
4	line 4 `x = 2;`	x changes to 2. The snapshot inside **byVal** is still 1. **byRef** sees the change.	**`x = 2`** | `byVal.x = 1` | `byRef.x = alias`
5	line 5 `std::cout << byVal() << byRef();`	Prints **12**. A reference capture that outlives x (returned, or run later on another thread) **dangles**: capture by value when a lambda escapes its scope. prints: `12\n`	`x = 2` | `byVal.x = 1` | `byRef.x = alias`

DANGLING CAPTURES A lambda that captures a local by reference and then outlives it (returned from the function, or run later on another thread) reads destroyed memory. When a lambda escapes its scope, capture by value.

[=] AND THIS Inside a member function `[=]` also captures `this` (the pointer), so the lambda uses the object through that pointer and dangles if the object dies. This implicit capture is deprecated in C++20: write `[=, this]` to say so, or `[*this]` to copy the object.

READ MORE

C++ Primer (Lippman, Lajoie, Moo)

- 10.3 Customizing operations (PDF pages 414 to 429)

`mutable`, init-capture, generic and recursive lambdas

```
auto count = [n = 0]() mutable { return ++n; };           // own state;
mutable lets it change
count(); count(); count();                                // 1 2 3 (compiled:
"123")

auto owned = std::make_unique<int>(42);
auto init  = [p = std::move(owned)] { return *p; };       // C++14 init-
capture: MOVE a move-only object in
                                                          // (compiled:
returns 42, the original is null)

auto bigger = [](const auto& a, const auto& b) { return a < b ? b : a; };
// C++14 generic: any type
bigger(3, 7);  bigger(2.5, 1.5);                          // 7 and 2.5

auto result = [&] { return compute(a, b); }();            // immediately
invoked: initialise a const with logic
const int limit = [&] { return fast ? 100 : 10; }();
```

- **Init-capture** `[name = expr]` creates a new member: the way to move a `unique_ptr` in, or to precompute something.
- `mutable` changes the lambda's own copy; each copy of the lambda has its own state.
- A **generic lambda** has a templated `operator()`: `auto` parameters are template parameters.
- A lambda cannot name itself. For recursion, pass it to itself: `auto fact = [](auto self, int n) -> int { return n <= 1 ? 1 : n * self(self, n - 1); };` (compiled: 5! = 120), or use `std::function`.
- `[this]` in a member function captures the pointer; `[*this]` copies the object.

Function pointer, functor, lambda, `std::function`

Four ways to pass "something callable":

KIND	CAN HOLD STATE?	INLINABLE?	OVERHEAD
function pointer	no	rarely (indirect call)	8 bytes
functor (class with `operator()`)	yes	yes: type known at compile time	size of its members
lambda	yes (captures)	yes: it is a functor	size of its captures
`std::function<R(Args...)>`	yes	no: type-erased	indirect call, and may allocate

```
template <typename F>
void apply(F f) { f(1); }                                   // template parameter: the lambda's exact type, fully inlined

void apply2(std::function<void(int)> f) { f(1); }           // type-erased: accepts anything, one indirect call

std::vector<std::function<int(int)>> ops{ [](int x) { return x + 1; },
                                          [k = 10](int x) { return x * k; }
};   // different lambdas, one type
```

- **Templates for hot paths and algorithms** (`std::sort` takes its comparator as a template parameter and inlines it). `std::function` when you must **store** callables of different types or keep them behind an interface.
- `std::function` may **allocate** when the callable does not fit its small buffer. Compiled with libstdc++: a small capture did not allocate, a 64-byte capture did.
- `std::invoke(f, args...)` calls anything callable, including a member function pointer: `std::invoke(&Account::balance, acct)`.
- A stateless lambda converts to a function pointer, handy for C APIs and callbacks.

`std::bind`: know it, prefer a lambda

```
using namespace std::placeholders;
auto f = std::bind(add3, 1, _1, _2);     // fix the first argument, leave two
open: f(2, 3) is add3(1, 2, 3)
auto g = std::bind(add3, _2, _1, 9);     // placeholders can reorder: g(4, 5)
is add3(5, 4, 9)
auto h = [](int b, int c) { return add3(1, b, c); };   // the same as f, and
easier to read
```

Compiled: `f(2, 3)` gave 123, `g(4, 5)` gave 549 and the lambda gave 123. `std::bind` predates lambdas (C++11); it copies its arguments, hides overloads and produces unreadable errors. You will see it in older code, so be able to read it; in new code write a lambda, which also inlines better.

KEY POINT "Lambda vs `std::function`?" A lambda has its own type and is inlined; `std::function` erases the type, costs an indirect call and maybe an allocation. Use it only when you must store different callables in one place.

READ MORE

C++ Primer (Lippman, Lajoie, Moo)

- 14.8 The function-call operator (PDF pages 600 to 607)

Cheat sheet

Answer aloud, then open.

▼ **What is a lambda and what does the compiler generate?**

An unnamed function object: the compiler creates a unique class with the captured variables as members and the body as `operator()`. Because it has its own type, calls can be inlined.

▼ Capture by value vs by reference?

By value copies the variable when the lambda is created (const inside unless `mutable`). By reference stores a reference to the live variable and dangles if the lambda outlives it. Capture by value for lambdas that escape the scope or run on another thread.

▼ How do you move a `unique_ptr` into a lambda?

With a C++14 init-capture: `[p = std::move(ptr)] { ... }`. A plain capture would need a copy, which `unique_ptr` forbids.

▼ What does `mutable` do on a lambda?

It makes `operator()` non-const so the lambda can modify its by-value captures. The changes live in the lambda's own copy, not in the original variable.

▼ `std::function` vs a template parameter?

A template parameter keeps the exact callable type, so the call is inlined and nothing allocates. `std::function` erases the type: one indirect call, possible allocation, but it can store different callables behind one type.

▼ What is a generic lambda?

A lambda with `auto` parameters (C++14): its `operator()` is a template, so one lambda works for any argument type.

▼ What is wrong with capturing `this` by `[=]`?

`[=]` inside a member function captures the `this` pointer, not a copy of the object, so the lambda dangles if the object is destroyed first. It is deprecated in C++20: write `[=, this]` or `[*this]` to copy.